A counterexample to the Riordan nullity conjecture

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Task 1. Resolve the nullity equality conjecture. Conjecture 7.9 of Cheon, Jung, Kitaev and Mojallal [1] asserts that every singular io-decomposable Riordan graph of the Bell type has $\eta(A_n) = \eta(B)$, where, after listing odd vertices first,

$$A_n \sim \begin{pmatrix} A_{\lceil n/2 \rceil} & B \\ B^T & 0 \end{pmatrix}.$$

Here nullity is the dimension of the right kernel. In particular, B has $\lfloor n/2 \rfloor$ columns. We give a counterexample with $n = 15$, $\eta(A_{15}) = 1$, and $\eta(B) = 0$.

For clarity, the Bell-type graph $G_n(g, zg)$ has vertices $1, \ldots, n$, zero diagonal, and, for $i > j$, adjacency entry

$$a_{ij} = a_{ji} = [z^{i-j-1}]g(z)^j \pmod 2 \quad \text{in } \{0, 1\}. \tag{1}$$

It is io-decomposable when its even vertices form an independent set and its odd induced subgraph, relabelled in order, is $G_{\lceil n/2 \rceil}(g, zg)$.

Theorem 1. *Let*

$$E(z) = 1 + z^6 + z^8 + z^{12}, \qquad g(z) = \sum_{s=0}^{\infty} z^{2^s-1} E(z^{2^s}).$$

Then $G_{15}(g, zg)$ is a singular, proper, io-decomposable Riordan graph of the Bell type with $\eta(A_{15}) = 1 \neq 0 = \eta(B)$.

Proof. The displayed series is a well-defined element of $\mathbb{Z}[[z]]$ with $g(0) = 1$. Thus the Riordan pair is proper. It satisfies $g(z) = E(z) + zg(z^2)$. Working modulo 2, its odd part is $zg(z^2)$, and $g(z)^2 = g(z^2)$. For $i > j$, the even-even adjacency entry is

$$[z^{2(i-j)-1}]g(z)^{2j} = 0.$$

The odd part of $g(z)^{2j-1} = g(z)(g(z)^{j-1})^2$ is $zg(z^2)^j$. Consequently

$$a_{2i-1,2j-1} = [z^{2(i-j)-1}]g(z)^{2j-1} = [z^{i-j-1}]g(z)^j = a_{ij} \pmod 2.$$

This proves io-decomposability for every order, including 15.

Only the following initial coefficients are needed for A_{15}:

$$g(z) \equiv 1 + z + z^3 + z^6 + z^7 + z^8 + z^{12} + z^{13} \pmod{z^{14}}.$$

Using (1), in the vertex order $(1, 3, 5, 7, 9, 11, 13, 15; 2, 4, 6, 8, 10, 12, 14)$ its blocks are

$$C = A_8 = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Let

$$o = (1, 0, 1, 0, -1, 0, -1, -1)^T, \qquad e = (-1, 0, -1, -1, 2, -1, 1)^T.$$

Direct integer multiplication gives

$$B^T o = 0, \qquad Co = (-1, 1, 0, 0, 1, -1, -2, 0)^T = -Be.$$

Thus $(o, e)^T \neq 0$ is in the kernel of the reordered adjacency matrix. In the original vertex order it is

$$(1, -1, 0, 0, 1, -1, 0, -1, -1, 2, 0, -1, -1, 1, -1)^T.$$

We also give a short exact certificate that B has full column rank. Write b_i for row i of B, and write e_j^T for the standard row vectors in $\mathbb{R}^7$. Successively, the rows $b_2, \ldots, b_8$ span

$$\begin{aligned}
&e_4^T = b_4 - b_6, && e_3^T = b_6 - b_7, && e_2^T = b_2 - b_8 - e_3^T, \\
&e_7^T = b_7 - b_3 + b_2, && e_1^T = b_8 - e_7^T, \\
&e_5^T = b_5 - e_1^T - e_2^T - e_3^T - e_4^T, && e_6^T = b_3 - b_2 - e_5^T.
\end{aligned}$$

Hence $\operatorname{rank} B = 7$ and $\eta(B) = 0$. As an additional check, deleting the first row of B gives a square matrix of determinant -1.

Finally, projection of $\ker A_{15}$ onto its odd coordinates is injective: if those coordinates vanish, the other block equation is $Be = 0$, which forces $e = 0$. Its image lies in $\ker B^T$, which has dimension $8 - 7 = 1$. The exhibited nonzero kernel vector therefore proves $\eta(A_{15}) = 1$. All hypotheses of Conjecture 7.9 hold and its equality fails. □

Scope. This refutes the full universal conjecture by a concrete graph in the intended class. The series is specified to all orders; an arbitrary finite coefficient list is not assumed to extend to that class. The example also has nonzero odd coordinates in a zero eigenvector, contrary to the equivalent Conjecture 7.7. That equivalent formulation receives no additional campaign credit.

Task 2. Classify counterexamples of order fifteen. Determine all generating-series prefixes giving counterexamples of order fifteen and classify the resulting graphs up to isomorphism.

Task 3. Settle the even-order restriction. Determine whether $\eta(A_n) = \eta(B)$ holds for every even-order io-decomposable Bell-type Riordan graph, or exhibit an even-order counterexample.

Task 4. Characterize families with nullity equality. Use the restriction of the odd adjacency quadratic form to $\ker B^T$ to identify conditions on g that force nullity equality at every order.

Task 5. Construct an infinite family. Find an explicit family of distinct orders with counterexamples, tracking how the nonzero odd kernel coordinates propagate under the recursive odd-vertex decomposition.

References

[1] G.-S. Cheon, J.-H. Jung, S. Kitaev and S. A. Mojallal, *Riordan graphs II: Spectral properties*, arXiv:1801.07021; *Linear Algebra and its Applications* **575** (2019), 174–215, doi:10.1016/j.laa.2019.04.011. Definitions 2.2 and 2.9, Theorem 7.5, and Conjecture 7.9 appear on pp. 3, 6, 27, and 29 of the inspected 33-page arXiv and accepted author manuscripts.